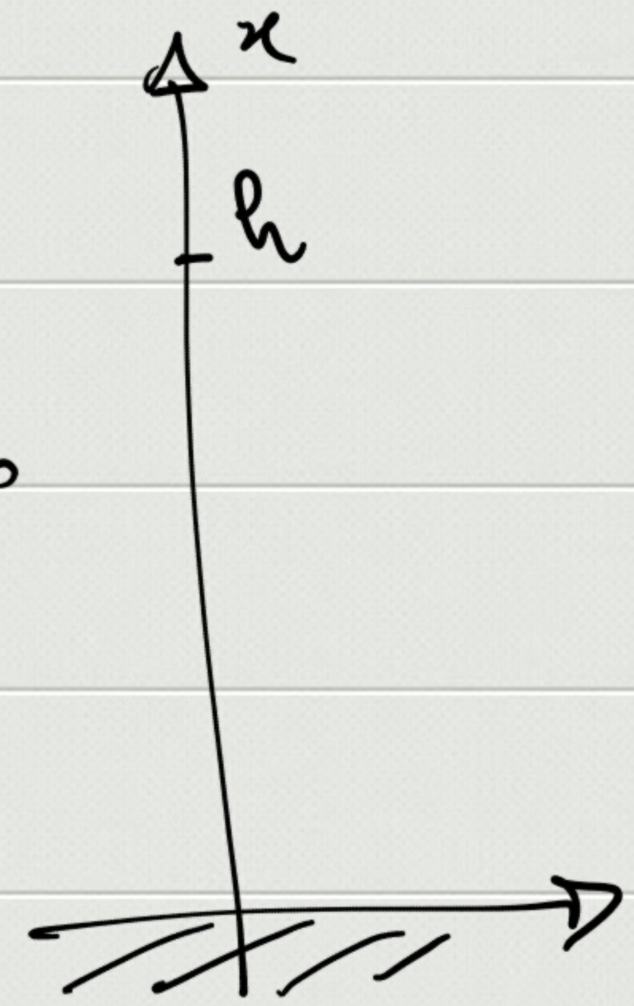
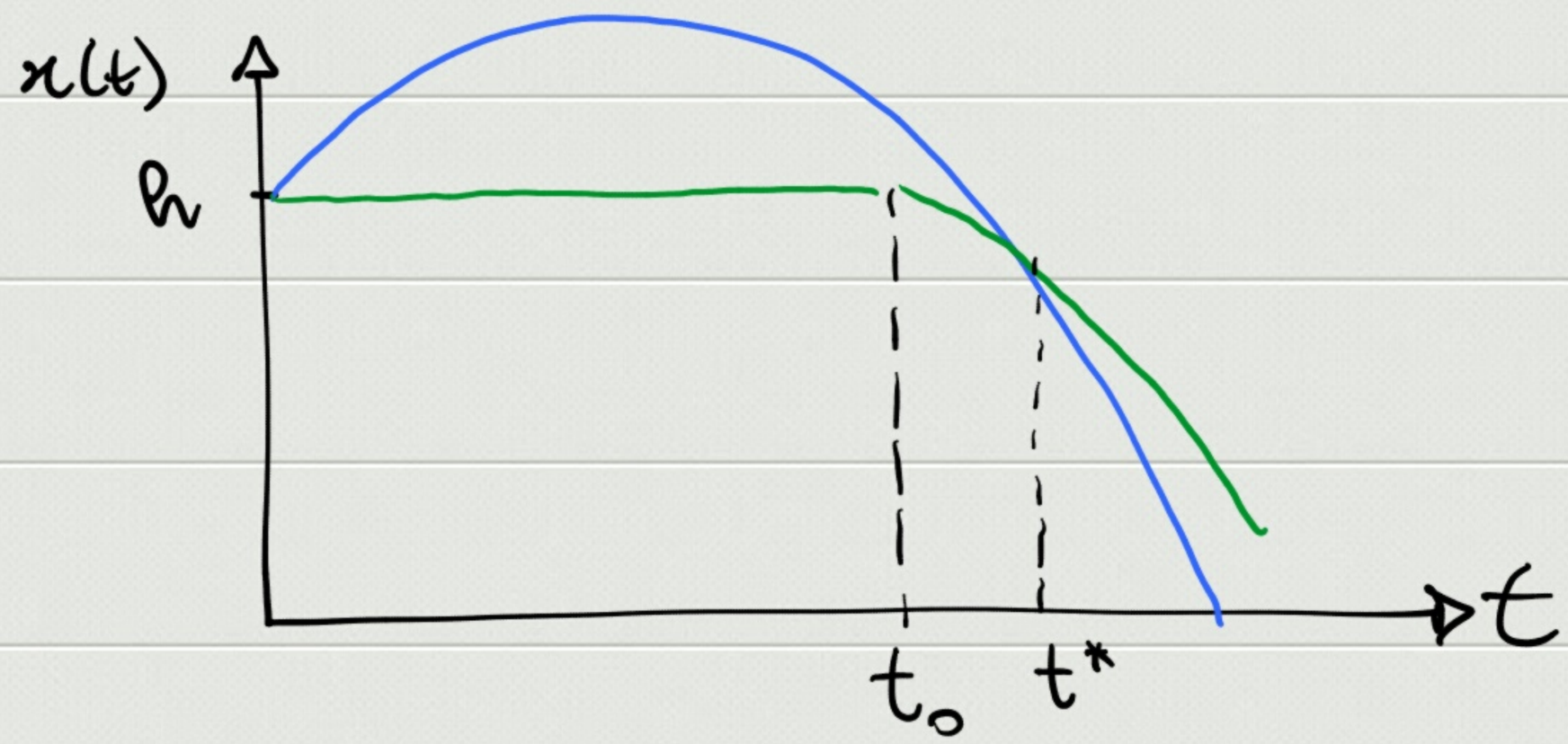
$x_{01} = x_{02} = h > 0$

$v_{01} = v_0 > 0 \qquad v_{02} = 0 \qquad t_{02} = t_0$



① $x_1(t) = h + v_0 t - \frac{1}{2} g t^2$ *

② $x_2(t) = h - \frac{1}{2} g (t - t_0)^2$ *



$x_1(t^*) = x_2(t^*)$

$$\cancel{h} + v_0 t^* - \frac{1}{2} g t^{*2} = \cancel{h} - \frac{1}{2} g (t^* - t_0)^2$$

$$v_0 t^* - \cancel{\frac{1}{2} g t^{*2}} = -\cancel{\frac{1}{2} g t^{*2}} + g t^* t_0 - \frac{1}{2} g t_0^2$$

$$t^* (v_0 - g t_0) = -\frac{1}{2} g t_0^2 \Rightarrow \boxed{t^* = \frac{g t_0^2}{2(g t_0 - v_0)}}$$

$$t^* = \frac{g t_0^{\cancel{2}}}{2(g t_0 - v_0)} > \cancel{t_0}^{1}$$

$$g t_0 - v_0 > 0 \left(\Rightarrow t_0 > \frac{v_0}{g}\right) \Rightarrow g t_0 > 2(g t_0 - v_0)$$

$$g t_0 < v_0 \cdot 2 \Rightarrow t_0 < \frac{2 v_0}{g}$$

$$\boxed{\frac{v_0}{g} < t_0 < \frac{2 v_0}{g}}$$

$$g t_0 - v_0 < 0 \left(\Rightarrow t_0 < \frac{v_0}{g}\right) \Rightarrow g t_0 < 2(g t_0 - v_0)$$

$$\Rightarrow g t_0 > 2 v_0 \quad t_0 > \frac{2 v_0}{g}$$